(Please write your Enrollment Number) Enrollment No. 0 32011 99023

End-Term Examination

(CBCS)(SUBJECTIVE TYPE)(OffLine)

Course Name: B.Tech , Semester: 3rd Sem.

(December, 2024)

Subject Code: BCS 203	Subject: Discrete Structures
Time :3 Hours	Maximum Marks : 60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (5*4=20)

a) Identify the nature of the proposition S, whether it is Tautology/ Contingency/ Contradiction.
S: $((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$

b) Draw the Hasse Diagram of the following: D_{105}, and D_{72}

c) Prove that group G is an Abelian group if and only if $(ab)^{-1} = a^{-1}b^{-1}$, $\forall\, a, b \in G$.

d) The chromatic number of the following graph is _____

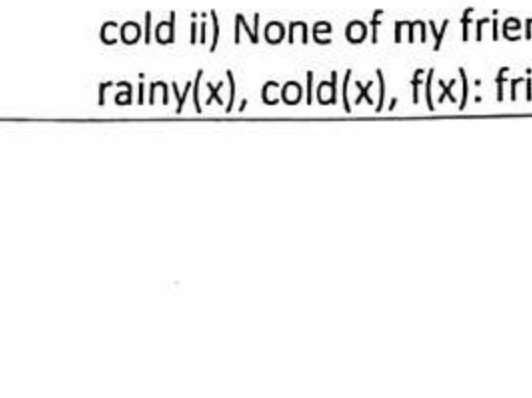


Fig. (a)

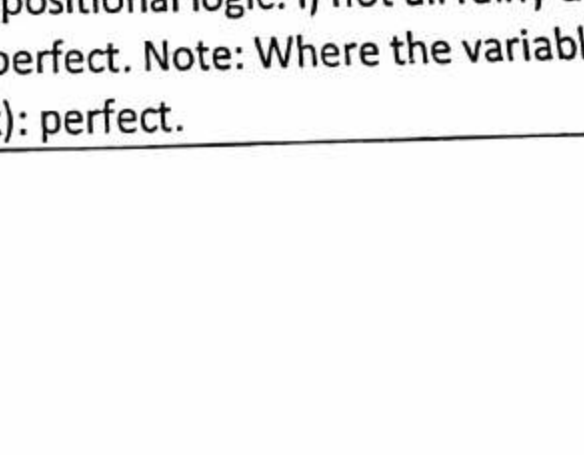


Fig. (b)

UNIT I

Q2 (5+5)

a) Let p, q, r, s represent the following propositions.
p: $x \in \{8,9,10,11,12\}$
q: x is a composite number
r: x is a perfect square
s: x is a prime number
The integer $x \geq 2$ which satisfies: $\neg((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is _____.

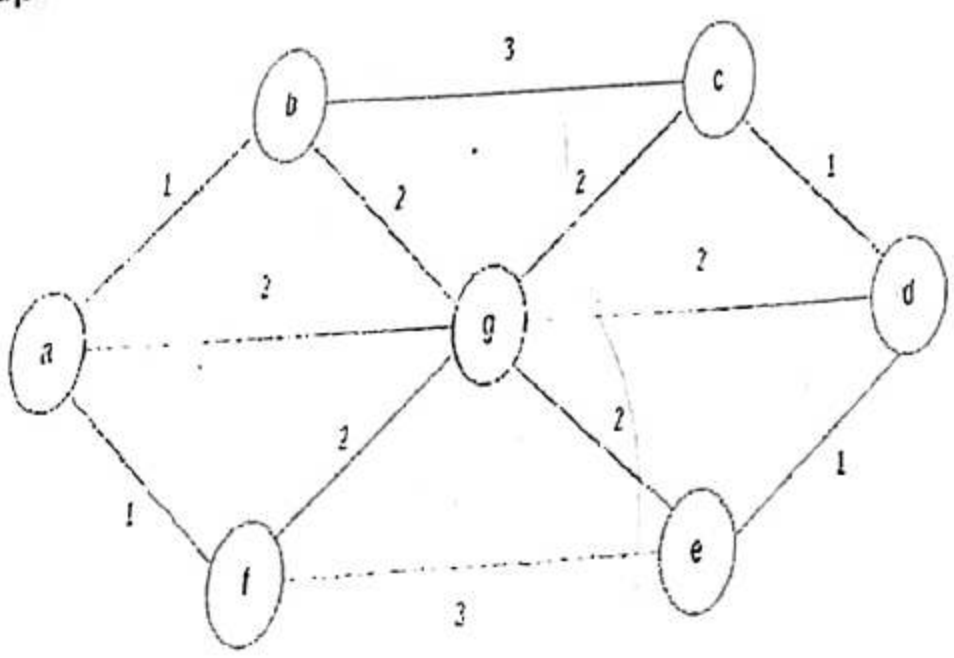
b) The binary operator $\neq$ is defined by the following truth table.

p	q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

Identify the nature of the binary operator $\neq$, whether it is associative, commutative, or both?

Q3 (5+5)

a) Translate the following into propositional logic: i) not all rainy days are cold ii) None of my friends are perfect. Note: Where the variables are: rainy(x), cold(x), f(x): friend, p(x): perfect.

	b) Identify the following Boolean expressions which is/are NOT tautology? A. $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$ B. $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$ C. $(a \wedge b \wedge c) \rightarrow (c \vee a)$ D. $a \rightarrow (b \rightarrow a)$	
	UNIT II	
Q4	a) Using the principle of mathematical induction, Show that 2^{2n}- 1 is divisible by 3. b) Find the total number of relation on a set R with n elements which is antisymmetric but not reflexive.	(5+5)
Q5	a) Prove that the relation congruence modulo m on the set Z of all integers is an equivalence relation. b) Explain the following Sets with example: a) Finite b) Infinite, c) Countable d) Uncountable.	()
	UNIT III	
Q6	a) Define a Field. Prove that the set of integers Z_{11} with addition and multiplication is a Field. b) State and prove Lagrange's theorem for finite groups.	(5+5)
Q7	a) Prove that a group of prime order p is cyclic. b) Let G be the set of all positive rational numbers and * be the binary operation on G defined as $a * b = \frac{ab}{7}, \forall a, b \in G$. Prove that $(G,*)$ be an abelian group.	(5+5)
	UNIT IV	
Q8	a) State and prove Euler's formula for connected planar graphs: V–E+R= 2, where V, E, and F represent vertices, edges, and region, respectively. b) The number of distinct minimum-weight spanning trees in the following graph is 	(5+5)
Q9	a) $G = (V,E)$ is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G. Which of the following statements about the minimum spanning trees (MSTs) of G is/are TRUE? I. If e is the lightest edge of some cycle in G, then every MST of G includes e. II. If e is the heaviest edge of some cycle in G, then every MST of G excludes e. b) Prove that: Any planar graph can be colored using at most five colors such that no two adjacent vertices share the same color.	(5+5)